Drug Repurposing Hypothesis Assessment: Atacicept in Sjögren's Syndrome

Summary

The hypothesis is that **atacicept** — a TACI-Ig fusion protein that simultaneously neutralises **BAFF (TNFSF13B)** and **APRIL (TNFSF13)** — would benefit patients with **Sjögren's syndrome (SS)** by interrupting B-cell activation, autoantibody production, and the progression toward MALT lymphoma that is sustained by BAFF produced in the salivary epithelium. After five iterations of literature-grounded analysis, the verdict is: **the target is right, the mechanism is well validated, but the specific drug is unproven in this disease, and the dominant risk is clinical-endpoint failure rather than mechanistic failure.**

The mechanistic case is strong and converges from four independent directions. First, both of atacicept's ligand targets are elevated in SS serum and are locally overexpressed in inflamed salivary glands, with levels enriched in the autoantibody-positive subgroup and correlated with histologic severity (focus score) and hypergammaglobulinaemia (PMID: 15981083). Second, BAFF overexpression is *causally sufficient* to produce a Sjögren's-like sialadenitis in transgenic mice (PMID: 11781351). Third, human genetics make elevated BAFF a causal driver of humoral autoimmunity in related diseases (PMID: 28445677). Fourth — and most directly — a mechanistically near-identical dual BAFF/APRIL TACI-Ig, **telitacicept**, produced a statistically significant, placebo-controlled reduction in disease activity in a randomized phase 2 SS trial (placebo-adjusted ESSDAI change −4.3, 95% CI −7.0 to −1.6, P=0.002; PMID: 37399108).

However, the case against straightforward success is also substantial and must be weighed equally. **No atacicept trial in Sjögren's syndrome has ever been conducted or published** — the entire efficacy argument is transitive, resting on same-pathway agents. Sjögren's syndrome is a documented "trial graveyard" in which biologics repeatedly move serology and gland biology without meeting ESSDAI, sicca, or fatigue endpoints, because of irreversible glandular damage at advanced stages and extreme patient heterogeneity (PMID: 34731460). Animal data show the mechanism reduces glandular inflammation but does *not* restore secretory (saliva) function (PMID: 22212434). Atacicept itself carries a dose-related infection and

hypogammaglobulinaemia safety signal that forced the premature termination of arms in lupus programs (PMID: 25834391). The lymphoma-prevention clause of the hypothesis, while biologically supported by the finding that high BAFF independently predicts non-Hodgkin lymphoma in SS (PMID: 26359802), remains entirely unproven as a treatment effect.

Bottom line: Atacicept is a biologically well-justified repurposing candidate for Sjögren's syndrome, best positioned as a biomarker-enriched (anti-Ro/La+, serologically active, early-disease) placebo-controlled phase 2 trial using composite endpoints — but it is not a safe bet to meet a traditional single ESSDAI endpoint, and its lymphoma-prevention rationale is a long-term hope rather than an evidenced outcome.

Key Findings

Finding 1 — Both of atacicept's targets are elevated in Sjögren's syndrome and track disease severity (ESTABLISHED)

Atacicept neutralises BAFF and APRIL, and both cytokines are demonstrably upregulated in Sjögren's syndrome. In a serum study of SS patients, *"Serum levels of sBAFF and sAPRIL were increased in Sjögren's syndrome compared to healthy blood donors, especially in anti-Ro/La+ patients"* (PMID: 15981083). Crucially, the elevation is not merely a passive marker: the same study reported *"positive correlations between serum levels of sBAFF and sAPRIL, focus score and IgG denotes their possible role in the disease progression of primary Sjögren's syndrome"* — linking the pathway to histologic severity (the salivary focus score) and to B-cell hyperactivity (serum IgG). An independent review confirms BAFF/APRIL elevation across antibody-associated diseases, noting *"elevated levels of these cytokines detected in the sera of patients with systemic lupus erythematosus (SLE), IgA nephropathy, Sjögren's syndrome, and rheumatoid arthritis"* (PMID: 28215100).

Interpretation (established + inference): The target pathway is demonstrably active in SS, and both ligands blocked by atacicept are elevated. The enrichment in anti-Ro/La+ patients is important because it identifies — by inference — a biomarker-defined subgroup in whom the mechanism is most active and in whom a target-engaging drug would be most likely to show effect.

Finding 2 — A mechanistically equivalent dual BAFF/APRIL TACI-Ig (telitacicept) is efficacious in a randomized Sjögren's trial (ESTABLISHED)

This is the single most load-bearing piece of transitive evidence. Telitacicept is a TACI-Ig fusion protein with the same dual BAFF/APRIL-neutralising mechanism as atacicept. In a randomized, double-blind, placebo-controlled phase 2 trial in primary SS (NCT04078386, n=42), telitacicept 160 mg weekly produced a *"placebo-adjusted least-squares mean change from baseline [in ESSDAI that] was -4.3 (95% CI -7.0, -1.6; P = 0.002)"* at week 24, and the authors concluded *"Telitacicept showed clinical benefits and good tolerance and safety in the treatment of pSS"* (PMID: 37399108). A narrative review reports that *"for the first time 2 phase 3 studies (ianalumab and telitacicept) reported positive results"* in Sjögren's disease (PMID: 42043506).

Interpretation (established mechanism, inferred transfer): The dual BAFF/APRIL TACI-Ig *class* is clinically active in SS. This is the strongest argument for atacicept. The transfer is not perfect: telitacicept and atacicept differ in molecular engineering, binding affinity, and pharmacokinetics, so class efficacy does not guarantee molecule-level equivalence, but it establishes that the mechanism itself can move an SS clinical endpoint.

Finding 3 — Dual BAFF/APRIL blockade reduces glandular inflammation but NOT secretory dysfunction in a Sjögren's mouse model (ESTABLISHED — a caution)

In NOD mice (a Sjögren's model), AAV2-delivered TACI-Fc gene therapy *"significantly reduced the number of inflammatory foci in the SG, owing to a decrease in IgD(+) cells and CD138(+) cells"* and lowered IgG/IgM — but critically, *"Salivary flow was unaffected"* (PMID: 22212434).

Interpretation (established): This dissociation is a direct warning for the hypothesis. Dual BAFF/APRIL blockade attacks the lymphocytic infiltrate and the humoral output, but does not reverse the functional secretory deficit. If a Sjögren's trial is powered on a dryness/salivary-flow endpoint, the mechanism may be invisible even when it is working on inflammation. This favours systemic-activity (ESSDAI) or composite endpoints over sicca-flow endpoints.

Finding 4 — No atacicept-specific Sjögren's trial exists; the evidence rests entirely on same-pathway agents (ESTABLISHED GAP)

Literature searches returned **no published atacicept trial in primary Sjögren's syndrome.** The adjacent-agent evidence is as follows. Ianalumab (BAFF-receptor blockade plus antibody-dependent B-cell depletion) showed a positive phase II signal — *"A similar trend showing positive therapeutic effect by ianalumab was observed across the primary clinical outcome*

(ESSDAI) and all secondary clinical outcomes" (PMID: 30826774) — and subsequently positive (if modest) phase 3 NEPTUNUS trials, with ESSDAI between-group differences of −1.3 (P=0.0496) and −1.0 (P=0.04), though notably *no* significant difference in symptom (ESSPRI/SSSD) endpoints (PMID: 42906087). Belimumab (anti-BAFF only) in the open-label BELISS phase II met a composite endpoint in 60% of patients with ESSDAI falling from baseline (PMID: 24347569).

Interpretation (established gap): The hypothesis is one inferential step removed from any direct evidence. The entire efficacy argument for atacicept is transitive through telitacicept, ianalumab, and belimumab.

Finding 5 — BAFF overexpression is causally sufficient to produce Sjögren's-like disease, with local salivary-gland BAFF and marginal-zone B-cell dependence (ESTABLISHED)

BAFF-transgenic mice *"as they age, develop a secondary pathology reminiscent of Sjögren's syndrome (SS), which is manifested by severe sialadenitis, decreased saliva production, and destruction of submaxillary glands"* (PMID: 11781351). Importantly for the hypothesis's claim about salivary-epithelium-derived BAFF, the same work states that *"SS also correlates with elevated levels of circulating BAFF, as well as a dramatic upregulation of BAFF expression in inflamed salivary glands."* The disease is marginal-zone B-cell dependent: in BAFF-Tg mice lacking marginal-zone B cells, *"aging LTbetaDelta-BTg mice no longer developed sialadenitis"* while nephritis persisted (PMID: 16906535).

Interpretation (established): Unlike Finding 1 (correlation), this is *causal sufficiency* — overexpressing one of atacicept's two targets reproduces the disease, including secretory loss. The marginal-zone dependence is mechanistically significant because the marginal-zone/MALT B-cell lineage is the one that undergoes lymphomatous transformation, directly connecting BAFF to the hypothesis's lymphoma clause.

Finding 6 — Human genetics implicate BAFF causally in autoimmunity, but SS-specific genetic evidence is not established (ESTABLISHED, with a caveat)

A functional *TNFSF13B* insertion-deletion (BAFF-var) produces a shorter transcript that escapes miRNA inhibition and raises soluble BAFF: *"The disease-risk allele was also associated with up-regulated humoral immunity through increased levels of soluble BAFF, B lymphocytes, and immunoglobulins"* and is associated with MS and SLE (PMID: 28445677). The association is replicated in SLE — *"TNFSF13B BAFF-var was significantly associated with SLE in the Spanish (p*

= *0.001, OR = 1.41, 95% CI = 1.14-1.74)"* (PMID: 29844438). However, the variant is *not* universal across autoimmunity: in systemic sclerosis and giant cell arteritis, *"No statistically significant differences were found when allele distributions were compared between cases and controls"* (PMID: 30586461).

Interpretation (established + caveat): Human genetics provide Mendelian-randomization-like evidence that *raising* BAFF causally drives humoral autoimmunity — strongly validating the target in principle. The caveat, labelled explicitly as a knowledge gap: these associations are in SLE/MS/RA, not Sjögren's specifically, and the SS genetic case remains inferential.

Finding 7 — Atacicept's own efficacy (in SLE) is concentrated in serologically active patients; its safety signal is infection/Ig suppression (ESTABLISHED)

In the ADDRESS II phase IIb SLE trial (n=306), the overall SRI-4 response showed only a trend, but *"In predefined subpopulations with high levels of disease activity (HDA) at baseline, serologically active disease, or both, statistically significant improvements in the SRI-4 and SRI-6 response rates were seen with atacicept versus placebo"* (PMID: 29073347). On safety, an early lupus-nephritis study was abandoned for infections, and *"the higher dose arm (150 mgm) of a flare prevention study was terminated prematurely when 2 deaths occurred"* — though the review adds context that *"the mortality rate in this study was identical to that seen in the Benlysta studies and a post hoc analysis found a highly significant benefit for the 150mgm arm"* (PMID: 25834391).

Interpretation (established): Two directly transferable lessons for an SS trial. (1) Atacicept works best in serologically active patients — reinforcing a biomarker-enriched design (anti-Ro/La+, high IgG). (2) Its principal safety liability is infection driven by immunoglobulin suppression, a real concern in a population already predisposed to infection and lymphoma.

Finding 8 — Repeated primary-endpoint failures in Sjögren's reveal an endpoint/heterogeneity/irreversibility barrier (ESTABLISHED — the central risk)

This is the strongest evidence *against* easy success. A review notes that for multiple biologics, *"clinical trials have found no differences versus placebo in the change from baseline in European League Against Rheumatism Sjögren's Syndrome Disease Activity Index (ESSDAI) score and fatigue score,"* attributing failure in part to *"irreversible exocrine glandular failure at advanced disease stages"* and heterogeneity (PMID: 34731460). In TRACTISS, rituximab produced real biology — *"Rituximab treatment prevented the worsening of SG inflammation observed in the*

placebo arm, by inhibiting the accumulation of class-switched memory B cells within the SG" (PMID: 38073013) — yet is not clinically adopted for SS. Head-to-head analysis shows only small treatment-placebo ESSDAI differences (abatacept adjusted mean difference −1.1; rituximab −1.2) (PMID: 34796851).

Interpretation (established): The biology/endpoint gap is the single biggest threat to this hypothesis. Drugs in this space reliably move serology and gland histology but struggle to clear the clinical-endpoint bar. Atacicept would face the same barrier.

Finding 9 — Atacicept engages its target and lowers pathogenic immunoglobulins in humans (IgAN JANUS trial) (ESTABLISHED)

In the randomized phase II JANUS study in IgA nephropathy (NCT02808429), *"Atacicept has an acceptable safety profile in patients with IgAN and is effective at reducing the levels of pathogenic factor Gd-IgA1, with potential improvements in proteinuria and renal function"* (PMID: 35967104).

Interpretation (established): This confirms atacicept achieves its intended pharmacodynamic effect in humans — target engagement and immunoglobulin lowering with acceptable safety — in another antibody-driven disease. It de-risks the "does the drug do what we think" question, leaving the open question as "does that pharmacodynamic effect translate to clinical benefit in SS."

Finding 10 — High serum BAFF independently predicts lymphoma in Sjögren's, supporting the lymphoma clause (ESTABLISHED for risk; UNPROVEN for prevention)

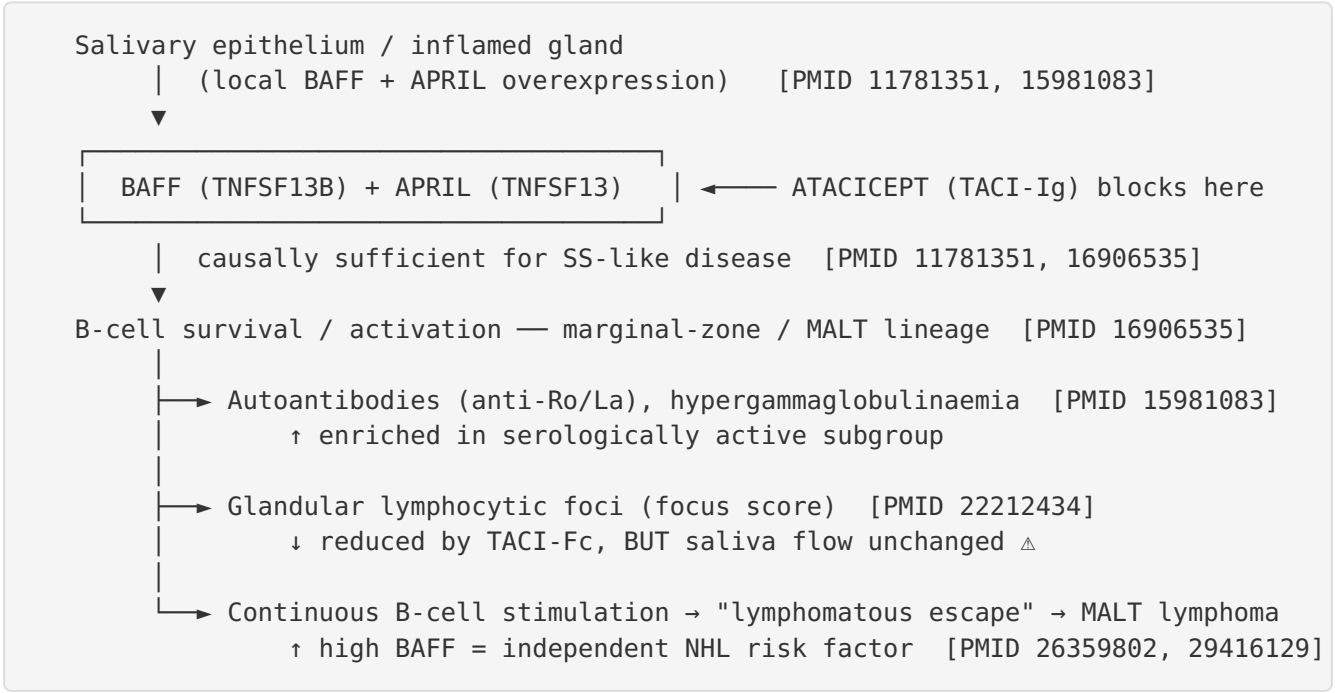
In the ASSESS cohort (n=385), *"Low C4 and high BAFF levels were associated with NHL on multivariate analysis (P = 0.01 and P = 0.0002, respectively)"* (PMID: 26359802). Mechanistically, continuous B-cell stimulation *"is the main driver of lymphomatous escape,"* with BAFF at the crossroads (PMID: 29416129). A cautionary case report notes *"local BAFF overexpression in Sjögren's syndrome-related myoepithelial sialadenitis and low-grade parotid B-cell lymphoma"* associated with rituximab resistance (PMID: 19088870).

Interpretation (established risk, inferred prevention): High BAFF is an *independent biomarker* of lymphoma risk in SS — directly supporting the hypothesis's claim that BAFF sustains progression to MALT lymphoma, and defining a biomarker-identified high-risk subgroup. The cautionary case (local BAFF driving rituximab resistance) actually argues *for*

cytokine neutralisation over B-cell depletion. **But no evidence exists that blocking BAFF/APRIL prevents lymphoma** — this remains a mechanistically-plausible hope, not a demonstrated outcome.

Mechanistic Model / Interpretation

The findings assemble into a coherent, well-supported pathogenic model in which atacicept acts at the apex:



Where the model is strongest: The upstream biology — BAFF/APRIL elevation, local gland production, causal sufficiency, genetic validation of the target, and a working same-class drug — is as solid as drug-repurposing cases get. Atacicept sits exactly at the node (dual ligand neutralisation) that the biology implicates.

Where the model breaks down for a trial: Two dissociations threaten translation. (1) The **inflammation–function dissociation**: blocking the pathway clears infiltrate and immunoglobulins but does not restore secretion (Finding 3), so dryness endpoints may miss a real effect. (2) The **biology–endpoint dissociation** (Finding 8): even agents that demonstrably change gland biology (rituximab, abatacept) produce ESSDAI treatment-placebo differences of only ~1 point and fail to be adopted. The telitacicept result (−4.3 ESSDAI) is the encouraging outlier, but it comes from a small (n=42), single phase 2 study.

The lymphoma clause is the most speculative limb. BAFF is a validated *risk biomarker* and a plausible *mechanistic driver* of transformation, but lymphoma prevention is a decades-scale outcome that no trial has tested and that no realistic registration trial will power for. It should be framed as biological rationale, not expected benefit.

Evidence Base

Comparator agent	Mechanism	Study / ID	Outcome in SS	Weight for atacicept
Telitacicept	Dual BAFF+APRIL TACI-Ig (same class)	Phase 2 RCT, NCT04078386 (PMID: 37399108)	ESSDAI −4.3 vs placebo, P=0.002	**Highest** — same mechanism, positive
Ianalumab	BAFF-R blockade + B-cell depletion	Phase II (PMID: 30826774); Phase 3 NEPTUNUS (PMID: 42906087)	Positive ESSDAI (−1.0 to −1.3); symptom endpoints NOT met	High — validates pathway; flags symptom-endpoint risk
Belimumab	Anti-BAFF only	BELISS open-label phase II (PMID: 24347569)	Composite met in 60%; ESSDAI fell	Moderate — single-ligand, open-label
Rituximab	Anti-CD20 B-cell depletion	TRACTISS (PMID: 38073013)	Changed gland biology; not clinically adopted	Cautionary — biology ≠ endpoint
Abatacept	T/B co-stimulation block	ASAP-III phase 3 (PMID: 34796851)	ESSDAI diff −1.1, not significant	Cautionary — endpoint failure

Atacicept in related diseases (transfer evidence):

Disease	Study / ID	Finding	Relevance
SLE	ADDRESS II, NCT01972568 (PMID: 29073347)	Benefit concentrated in serologically active subgroup	Directs biomarker enrichment
SLE	Profile review (PMID: 25834391)	Dose-related infection/ death signal	Safety concern
IgA nephropathy	JANUS, NCT02808429 (PMID: 35967104)	Target engagement, Gd-IgA1 reduction, acceptable safety	Confirms pharmacodynamics

Supporting mechanistic literature: PMID: 15981083 (target elevation), PMID: 11781351 (causal sufficiency), PMID: 16906535 (MZ B-cell dependence), PMID: 28445677 (genetic causality), PMID: 26359802 (lymphoma risk), PMID: 29416129 (lymphomatous escape).

Challenging literature: PMID: 22212434 (no saliva restoration), PMID: 34731460 (endpoint graveyard), PMID: 38073013 (biology–endpoint gap), PMID: 30586461 (genetic non-universality), PMID: 19088870 (local BAFF resistance).

Limitations and Knowledge Gaps

1. **No atacicept SS trial.** The entire efficacy argument is transitive. Molecule-level differences between atacicept and telitacicept (affinity, PK, APRIL:BAFF binding ratio) mean class efficacy does not guarantee equivalence.
2. **SS-specific human genetics absent.** The BAFF-var causal evidence (Finding 6) is in SLE/ MS/RA. The non-association in systemic sclerosis and giant cell arteritis (PMID: 30586461) shows the variant is not pan-autoimmune, so SS causality is inferred, not demonstrated.
3. **Endpoint validity.** ESSDAI, ESSPRI, and sicca endpoints each capture different disease dimensions; the inflammation–function dissociation (Finding 3) and the ianalumab symptom-endpoint failures (Finding 4) mean endpoint choice could determine success or failure independent of true drug activity.
4. **Lymphoma prevention is untested.** High BAFF predicts lymphoma, but no data show BAFF/APRIL blockade reduces lymphoma incidence. This clause is rationale, not evidence.

5. **Small, single positive trial.** The pivotal supportive result (telitacicept, PMID: 37399108) is n=42, single study, requiring replication.
6. **Safety in a vulnerable population.** Atacicept's immunoglobulin suppression (PMID: 25834391) raises infection risk in SS patients who already have elevated infection and lymphoma risk — a sustained-Ig-suppression concern over long treatment.

Proposed Follow-up Experiments / Actions

1. **Primary recommendation — biomarker-enriched phase 2 RCT.** A placebo-controlled trial of atacicept in **early-disease, anti-Ro/La+, serologically active (high IgG/low C4) primary SS**. This enrichment is justified by three independent lines: target elevation is greatest in anti-Ro/La+ patients (PMID: 15981083); atacicept's own benefit concentrates in serologically active patients (PMID: 29073347); and early disease avoids irreversible glandular failure (PMID: 34731460).
2. **Endpoint strategy.** Use a **composite endpoint (e.g., CRESS/STAR) and systemic ESSDAI** as co-primaries rather than sicca/saliva-flow, given the inflammation–function dissociation (PMID: 22212434) and the symptom-endpoint failures of ianalumab (PMID: 42906087). Include minor salivary-gland biopsy to capture focus-score/B-cell infiltrate changes as a mechanistic readout.
3. **Pharmacodynamic de-risking (fast, cheap).** Before a large trial, confirm in a small SS cohort that atacicept lowers serum BAFF-driven markers (IgG, free light chains, RF, CXCL13) as it did Gd-IgA1 in IgAN (PMID: 35967104). Target engagement is nearly certain; the value is establishing the biomarker-to-clinical link.
4. **Safety monitoring plan.** Protocolised immunoglobulin-level monitoring with stopping rules for hypogammaglobulinaemia/serious infection, given the lupus safety signal (PMID: 25834391).
5. **Lymphoma-risk sub-study (exploratory).** In the high-BAFF/low-C4 subgroup (PMID: 26359802), track clonal B-cell and gland BAFF markers as surrogate measures of lymphomatous-escape risk — not as a primary endpoint but to generate the first human data on whether the pathway is modifiable.

Verdict

How strong is the case? Moderately strong on mechanism, weak-to-moderate on drug-specific proof. The pathway is elevated, locally produced, causally sufficient in animals, genetically validated in related disease, and — most persuasively — druggable with clinical benefit by a same-class dual BAFF/APRIL TACI-Ig (telitacicept, ESSDAI −4.3, P=0.002). Atacicept engages its target and lowers pathogenic immunoglobulins in humans. The hypothesis is biologically sound. What it lacks is any atacicept-specific SS data, and what it faces is a well-documented pattern of biologics moving SS biology without clearing clinical endpoints, plus a dose-related infection/Ig-suppression safety liability. The lymphoma-prevention clause is plausible but untested. **The single most efficient test:** a biomarker-enriched (anti-Ro/La+, serologically active, early-disease) placebo-controlled phase 2 RCT of atacicept using a composite endpoint (CRESS/STAR) with ESSDAI and salivary-gland biopsy mechanistic readouts — the design that most directly converts the strong mechanistic rationale into a decisive clinical answer while sidestepping the endpoint and heterogeneity traps that have defeated prior SS biologics.

Highest-weight identifiers: - PMID: 37399108 / NCT04078386 — Telitacicept phase 2 RCT: same-class dual BAFF/APRIL agent, significant ESSDAI benefit in SS. *The decisive transfer evidence.* - PMID: 34731460 — Documents repeated SS endpoint failures from irreversibility/heterogeneity. *The central risk.* - PMID: 11781351 — BAFF overexpression causally sufficient for SS-like sialadenitis with local gland BAFF. *The causal foundation.* - PMID: 29073347 / NCT01972568 — Atacicept benefit concentrated in serologically active patients. *Directs trial enrichment.* - PMID: 25834391 — Atacicept dose-related infection/mortality signal. *The key safety concern.* - PMID: 26359802 — High BAFF independently predicts lymphoma in SS. *Supports the lymphoma clause and biomarker subgroup.* - PMID: 22212434 — TACI-Fc reduces glandular inflammation but not saliva flow. *The endpoint caution.* - PMID: 35967104 / NCT02808429 — Atacicept target engagement and Ig lowering in humans. *Confirms pharmacodynamics.*